

Finite Dimensional Vector Spaces (2005–2026 Archive)

Category: Linear Algebra and Algebra • Official Mock Test Series

TOTAL QUESTIONS

78

TOTAL MARKS

184 M

TEST DURATION

234 Min

PAPER PATTERN

MCQ • MSQ • NAT
(2005–20

Section / Type	Questions	Marking Scheme	Negative Marking	Format / Details
Multiple Choice (MCQ)	48	+1 / +2 Marks	−1/3 / −2/3 Marks	4 Options (1 Correct)
Multiple Select (MSQ)	10	+2 Marks	Nil (0)	4 Options (1+ Correct)
Numerical Answer (NAT)	20	+1 / +2 Marks	Nil (0)	Real Decimal Value

IMPORTANT INSTRUCTIONS FOR CANDIDATES

- Authentic Examination PYQs:** This mock paper contains verified official IIT JAM Mathematics questions and high-resolution problem statements.
- Section A (MCQ):** Each question has four choices (A), (B), (C), (D), out of which **only one** is correct. Wrong answers carry negative penalty (1/3 of positive marks).
- Section B (MSQ):** Each question has four choices (A), (B), (C), (D), out of which **one or more than one** choice(s) may be correct. Full marks are awarded only if all correct choices and zero incorrect choices are chosen. No partial marks and no negative marks.
- Section C (NAT):** Numerical answers are real numbers entered via virtual keyboard. Full credit is awarded if the numerical answer falls within the official accepted interval. No negative marking.
- Master Answer Key:** Complete official answers and acceptance bounds are provided at the end of this document.
- Online Interactive Simulator:** Practice this test with virtual calculator, timers, and instant scoring analytics at vaibhavgeometer.github.io.

IIT JAM Mathematics Mock Test Series • Curated by Vaibhav Geometer

Question 1 (JAM 2026 • Q40 • ID: JAM_2026_Q40)

MSQ

+2 M

Neg: Nil (0)

Q.40	Let V be the subset of $\mathbb{R}$ defined by $V = \left\{ \frac{a + b\sqrt{2}}{c + d\sqrt{2}} : a, b, c, d \in \mathbb{Q}, c^2 + d^2 \neq 0 \right\}.$ Which of the following statements is/are FALSE?
(A)	V is closed under the usual addition in $\mathbb{R}$.
(B)	V is a subspace of the real vector space $\mathbb{R}$.
(C)	V is a two dimensional subspace of the vector space $\mathbb{R}$ over $\mathbb{Q}$.
(D)	V is a four dimensional subspace of the vector space $\mathbb{R}$ over $\mathbb{Q}$.
Section C: Q.41 – Q.50 Carry ONE mark each.	

Question 2 (JAM 2026 • Q28 • ID: JAM_2026_Q28)

MCQ

+2 M

Neg: -0.67

Q.28	For any two points $P, Q \in \mathbb{R}^3$, let $\text{vect}(P, Q)$ denote the vector from the point P to the point Q and let $d(P, Q)$ denote the length of $\text{vect}(P, Q)$. Let $F: \mathbb{R}^3 \rightarrow \mathbb{R}^3$ be a linear transformation such that $d(F(P), F(Q)) = d(P, Q) \quad \text{for all } P, Q \in \mathbb{R}^3.$ Which ONE of the following statements is FALSE ?
(A)	F is an injective map.
(B)	F is a surjective map.
(C)	For any four points P, Q, R, S in $\mathbb{R}^3$, we have $\text{vect}(F(P), F(Q)) \cdot \text{vect}(F(R), F(S)) = \text{vect}(P, Q) \cdot \text{vect}(R, S),$ where $\cdot$ denotes the usual dot product in $\mathbb{R}^3$.
(D)	For any four points P, Q, R, S in $\mathbb{R}^3$, we have $\text{vect}(F(P), F(Q)) \times \text{vect}(F(R), F(S)) = \text{vect}(P, Q) \times \text{vect}(R, S),$ where $\times$ denotes the usual cross product in $\mathbb{R}^3$.

Question 3 (JAM 2026 • Q24 • ID: JAM_2026_Q24)

MCQ

+2 M

Neg: -0.67

Q.24

Let e_1, e_2, e_3, e_4, e_5 denote the standard basis vectors of the real vector space $\mathbb{R}^5$. Let V be the subspace of $\mathbb{R}^5$ spanned by the vectors $e_1 + e_2, e_2 + e_3, e_3 + e_4 + e_5$. Consider the real vector space $M_{2,5}(\mathbb{R})$. Let

$$S = \{P \in M_{2,5}(\mathbb{R}) \mid \text{nullspace}(P) = V\}.$$

Which ONE of the following statements is TRUE?

(A) S is the empty set.(B) $S = \{0\}$.(C) S is non-empty and S is not a subspace of $M_{2,5}(\mathbb{R})$.(D) S is a nonzero subspace of $M_{2,5}(\mathbb{R})$.

Question 4 (JAM 2025 • Q58 • ID: JAM_2025_Q58)

NAT

+2 M

Neg: Nil (0)

Q.58 Let the subspace H of $P_3(\mathbb{R})$ be defined as

$$H = \{p(x) \in P_3(\mathbb{R}) : xp'(x) = 3p(x)\}.$$

Then, the dimension of H is equal to _____.

Question 5 (JAM 2025 • Q49 • ID: JAM_2025_Q49)

NAT

+1 M

Neg: Nil (0)

Q.49 Let $T, S : P_4(\mathbb{R}) \rightarrow P_4(\mathbb{R})$ be the linear transformations defined by

$$T(p(x)) = xp'(x) \text{ and } S(p(x)) = (x+1)p'(x)$$

for all $p(x) \in P_4(\mathbb{R})$.

Then, the nullity of the composition $S \circ T$ is _____.

Question 6 (JAM 2025 • Q47 • ID: JAM_2025_Q47)

NAT

+1 M

Neg: Nil (0)

Q.47 Consider the real vector space $\mathbb{R}^3$. Let $T : \mathbb{R}^3 \rightarrow \mathbb{R}$ be a linear transformation such that

$$T(1, 1, 1) = 0, \quad T(1, -1, 1) = 0 \text{ and } T(0, 0, 1) = 16.$$

Then, the value of $T\left(\frac{1}{2}, \frac{2}{3}, \frac{3}{4}\right)$ is equal to _____ (rounded off to two decimal places).

Question 7 (JAM 2025 • Q46 • ID: JAM_2025_Q46)

NAT

+1 M

Neg: Nil (0)

Q.46 Consider the following subspaces of $\mathbb{R}^4$:

$$V_1 = \{(x, y, z, w) \in \mathbb{R}^4 : x + y + 2w = 0\},$$

$$V_2 = \{(x, y, z, w) \in \mathbb{R}^4 : 2y + z + w = 0\},$$

$$V_3 = \{(x, y, z, w) \in \mathbb{R}^4 : x + 3y + z + 3w = 0\}.$$

Then, the dimension of the subspace $V_1 \cap V_2 \cap V_3$ is equal to _____.

Question 8 (JAM 2025 • Q40 • ID: JAM_2025_Q40)

MSQ

+2 M

Neg: Nil (0)

Q.40 Consider the following subspaces of the real vector space $\mathbb{R}^3$:

$$V_1 = \text{span} \{(1, 2, 3), (1, 1, 0)\},$$

$$V_2 = \text{span} \{(1, -1, 0)\},$$

$$V_3 = \text{span} \{(1, 1, 1)\},$$

$$V_4 = \text{span} \{(1, 3, 6)\} \text{ and}$$

$$V_5 = \text{span} \{(1, 0, -3)\}.$$

Then, which of the following is/are TRUE?

(A) $V_1 \cup V_2$ is a subspace of $\mathbb{R}^3$

(B) $V_1 \cup V_3$ is a subspace of $\mathbb{R}^3$

(C) $V_1 \cup V_4$ is a subspace of $\mathbb{R}^3$

(D) $V_1 \cup V_5$ is a subspace of $\mathbb{R}^3$

Question 9 (JAM 2025 • Q35 • ID: JAM_2025_Q35)

MSQ

+2 M

Neg: Nil (0)

Q.35

Let

$$u_1 = (1, 0, 0, -1), u_2 = (0, 2, 0, -1), u_3 = (0, 0, 1, -1) \text{ and } u_4 = (0, 0, 0, 1)$$

be elements in the real vector space $\mathbb{R}^4$.

Then, which of the following is/are TRUE?

- (A) $\{u_1, u_2, u_3, u_4\}$ is a linearly independent set in $\mathbb{R}^4$
- (B) $\{u_1 - u_2, u_2 - u_3, u_3 - u_4, u_4 - u_1\}$ is NOT a linearly independent set in $\mathbb{R}^4$
- (C) $\{u_1, -u_2, u_3, -u_4\}$ is NOT a linearly independent set in $\mathbb{R}^4$
- (D) $\{u_1 + u_2, u_2 + u_3, u_3 + u_4, u_4 + u_1\}$ is a linearly independent set in $\mathbb{R}^4$

Question 10 (JAM 2025 • Q24 • ID: JAM_2025_Q24)

MCQ

+2 M

Neg: -0.67

Q.24 Let f_1, f_2, f_3 be nonzero linear transformations from $\mathbb{R}^4$ to $\mathbb{R}$ and

$$\ker(f_1) \subseteq \ker(f_2) \cap \ker(f_3).$$

Let $T : \mathbb{R}^4 \rightarrow \mathbb{R}^3$ be the linear transformation defined by

$$T(v) = (f_1(v), f_2(v), f_3(v)), \quad \text{for all } v \in \mathbb{R}^4.$$

Then, the nullity of T is equal to

(A) 1

(B) 2

(C) 3

(D) 4

Question 11 (JAM 2025 • Q8 • ID: JAM_2025_Q8)

MCQ

+1 M

Neg: -0.33

Q.8 Let $T : P_2(\mathbb{R}) \rightarrow P_2(\mathbb{R})$ be the linear transformation defined by $T(p(x)) = p(x + 1)$, for all $p(x) \in P_2(\mathbb{R})$. If M is the matrix representation of T with respect to the ordered basis $\{1, x, x^2\}$ of $P_2(\mathbb{R})$, then which one of the following is TRUE?

- (A) The determinant of M is 2
- (B) The rank of M is 2
- (C) 1 is the only eigenvalue of M
- (D) The nullity of M is 2

Question 12 (JAM 2025 • Q6 • ID: JAM_2025_Q6)

MCQ

+1 M

Neg: -0.33

Q.6 Define $T: \mathbb{R}^3 \rightarrow \mathbb{R}^3$ by $T(x, y, z) = (x + z, 2x + 3y + 5z, 2y + 2z)$, for all $(x, y, z) \in \mathbb{R}^3$.

Then, which one of the following is TRUE?

- (A) T is one-one and T is NOT onto
- (B) T is NOT one-one and T is onto
- (C) T is one-one and T is onto
- (D) T is NOT one-one and T is NOT onto

Question 13 (JAM 2024 • Q58 • ID: JAM_2024_Q58)

NAT

+2 M

Neg: Nil (0)

Q.58 For $k \in \mathbb{N}$, let $0 = t_0 < t_1 < \dots < t_k < t_{k+1} = 1$. A function $f: [0,1] \rightarrow \mathbb{R}$ is said to be piecewise linear with nodes $t_1, \dots, t_k$, if for each $j = 1, 2, \dots, k+1$, there exist $a_j \in \mathbb{R}$, $b_j \in \mathbb{R}$ such that

$$f(t) = a_j + b_j t \text{ for } t_{j-1} < t < t_j.$$

Let V be the real vector space of all real valued continuous piecewise linear functions on $[0,1]$ with nodes $\frac{1}{4}, \frac{1}{2}, \frac{3}{4}$. Then, the dimension of V equals

Question 14 (JAM 2024 • Q49 • ID: JAM_2024_Q49)

NAT

+1 M

Neg: Nil (0)

Q.49 Let $P_7(x)$ be the real vector space of polynomials in x with degree at most 7, together with the zero polynomial. For $r = 1, 2, \dots, 7$, define

$$s_r(x) = x(x-1)\dots(x-(r-1)) \text{ and } s_0(x) = 1.$$

Consider the fact that $B = \{s_0(x), s_1(x), \dots, s_7(x)\}$ is a basis of $P_7(x)$.

If

$$x^5 = \sum_{k=0}^7 \alpha_{5,k} s_k(x),$$

where $\alpha_{5,k} \in \mathbb{R}$, then $\alpha_{5,2}$ equals _____ (rounded off to two decimal places)

Question 15 (JAM 2024 • Q45 • ID: JAM_2024_Q45)

NAT

+1 M

Neg: Nil (0)

Q.45 Let $P_{12}(x)$ be the real vector space of polynomials in the variable x with real coefficients and having degree at most 12, together with the zero polynomial. Define

$$V = \left\{ f \in P_{12}(x) : f(-x) = f(x) \text{ for all } x \in \mathbb{R} \text{ and } f(2024) = 0 \right\}.$$

Then, the dimension of V is _____

Question 16 (JAM 2024 • Q22 • ID: JAM_2024_Q22)

MCQ

+2 M

Neg: -0.67

Q.22 Let $P_7(x)$ be the real vector space of polynomials, in the variable x with real coefficients and having degree at most 7, together with the zero polynomial. Let $T: P_7(x) \rightarrow P_7(x)$ be the linear transformation defined by

$$T(f(x)) = f(x) + \frac{df(x)}{dx}.$$

Then, which one of the following is TRUE?

- (A) T is not a surjective linear transformation
- (B) There exists $k \in \mathbb{N}$ such that T^k is the zero linear transformation
- (C) 1 and 2 are the eigenvalues of T
- (D) There exists $r \in \mathbb{N}$ such that $(T - I)^r$ is the zero linear transformation, where I is the identity map on $P_7(x)$

Question 17 (JAM 2024 • Q17 • ID: JAM_2024_Q17)

MCQ

+2 M

Neg: -0.67

Q.17 Let $P_{11}(x)$ be the real vector space of polynomials, in the variable x with real coefficients and having degree at most 11, together with the zero polynomial.

Let

$$E = \{s_0(x), s_1(x), \dots, s_{11}(x)\}, \quad F = \{r_0(x), r_1(x), \dots, r_{11}(x)\}$$

be subsets of $P_{11}(x)$ having 12 elements each and satisfying

$$s_0(3) = s_1(3) = \dots = s_{11}(3) = 0, \quad r_0(4) = r_1(4) = \dots = r_{11}(4) = 1.$$

Then, which one of the following is TRUE?

- (A) Any such E is not necessarily linearly dependent and any such F is not necessarily linearly dependent
- (B) Any such E is necessarily linearly dependent but any such F is not necessarily linearly dependent
- (C) Any such E is not necessarily linearly dependent but any such F is necessarily linearly dependent
- (D) Any such E is necessarily linearly dependent and any such F is necessarily linearly dependent

Question 18 (JAM 2024 • Q13 • ID: JAM_2024_Q13)

MCQ

+2 M

Neg: -0.67

Q.13 Let V be a nonzero subspace of the complex vector space $M_7(\mathbb{C})$ such that every nonzero matrix in V is invertible. Then, the dimension of V over $\mathbb{C}$ is

- (A) 1
- (B) 2
- (C) 7
- (D) 49

Question 19 (JAM 2023 • Q57 • ID: JAM_2023_Q57)

NAT

+2 M

Neg: Nil (0)

Q. 57 Let W be the subspace of $M_3(\mathbb{R})$ consisting of all matrices with the property that the sum of the entries in each row is zero and the sum of the entries in each column is zero. Then the dimension of W is equal to _____.

Question 20 (JAM 2023 • Q42 • ID: JAM_2023_Q42)

NAT

+1 M

Neg: Nil (0)

Q. 42 Let $T : P_2(\mathbb{R}) \rightarrow P_4(\mathbb{R})$ be the linear transformation given by $T(p(x)) = p(x^2)$. Then the rank of T is equal to _____.

Question 21 (JAM 2023 • Q39 • ID: JAM_2023_Q39)

MSQ

+2 M

Neg: Nil (0)

Q. 39 Which of the following is/are linear transformations?

- (A) $T : \mathbb{R} \rightarrow \mathbb{R}$ given by $T(x) = \sin(x)$
- (B) $T : M_2(\mathbb{R}) \rightarrow \mathbb{R}$ given by $T(A) = \text{trace}(A)$
- (C) $T : \mathbb{R}^2 \rightarrow \mathbb{R}$ given by $T(x, y) = x + y + 1$
- (D) $T : P_2(\mathbb{R}) \rightarrow \mathbb{R}$ given by $T(p(x)) = p(1)$

Question 22 (JAM 2023 • Q38 • ID: JAM_2023_Q38)

MSQ

+2 M

Neg: Nil (0)

Q. 38 Which of the following is/are true?

- (A) Every linear transformation from $\mathbb{R}^2$ to $\mathbb{R}^2$ maps lines onto points or lines
- (B) Every surjective linear transformation from $\mathbb{R}^2$ to $\mathbb{R}^2$ maps lines onto lines
- (C) Every bijective linear transformation from $\mathbb{R}^2$ to $\mathbb{R}^2$ maps pairs of parallel lines to pairs of parallel lines
- (D) Every bijective linear transformation from $\mathbb{R}^2$ to $\mathbb{R}^2$ maps pairs of perpendicular lines to pairs of perpendicular lines

Question 23 (JAM 2023 • Q15 • ID: JAM_2023_Q15)

MCQ

+2 M

Neg: -0.67

Q. 15 Let S and T be non-empty subsets of $\mathbb{R}^2$, and W be a non-zero proper subspace of $\mathbb{R}^2$. Consider the following statements:

- I. If $\text{span}(S) = \mathbb{R}^2$, then $\text{span}(S \cap W) = W$.
- II. $\text{span}(S \cup T) = \text{span}(S) \cup \text{span}(T)$.

Then

- (A) both I and II are TRUE
- (B) I is TRUE but II is FALSE
- (C) II is TRUE but I is FALSE
- (D) both I and II are FALSE

Question 24 (JAM 2023 • Q12 • ID: JAM_2023_Q12)

MCQ

+2 M

Neg: -0.67

Q. 12 Consider the following statements:

- I. There exists a linear transformation from $\mathbb{R}^3$ to itself such that its range space and null space are the same.
- II. There exists a linear transformation from $\mathbb{R}^2$ to itself such that its range space and null space are the same.

Then

- (A) both I and II are TRUE
- (B) I is TRUE but II is FALSE
- (C) II is TRUE but I is FALSE
- (D) both I and II are FALSE

Question 25 (JAM 2023 • Q3 • ID: JAM_2023_Q3)

MCQ

+1 M

Neg: -0.33

Q. 3 Which of the following is a subspace of the real vector space $\mathbb{R}^3$?

- (A) $\{(x, y, z) \in \mathbb{R}^3 : (y + z)^2 + (2x - 3y)^2 = 0\}$
- (B) $\{(x, y, z) \in \mathbb{R}^3 : y \in \mathbb{Q}\}$
- (C) $\{(x, y, z) \in \mathbb{R}^3 : yz = 0\}$
- (D) $\{(x, y, z) \in \mathbb{R}^3 : x + 2y - 3z + 1 = 0\}$

Question 26 (JAM 2021 • Q60 • ID: JAM_2021_Q60)

NAT

+2 M

Neg: Nil (0)

Q. 60 Let $A = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & -1 \end{pmatrix}$. Consider the linear map T_A from the real vector space $M_4(\mathbb{R})$ to itself defined by $T_A(X) = AX - XA$, for all $X \in M_4(\mathbb{R})$. The dimension of the range of T_A is _____.

Question 27 (JAM 2021 • Q46 • ID: JAM_2021_Q46)

NAT

+1 M

Neg: Nil (0)

Q. 46 Let V be the real vector space of all continuous functions $f : [0, 2] \rightarrow \mathbb{R}$ such that the restriction of f to the interval $[0, 1]$ is a polynomial of degree less than or equal to 2, the restriction of f to the interval $[1, 2]$ is a polynomial of degree less than or equal to 3 and $f(0) = 0$. Then the dimension of V is equal to _____.

MA

14 / 17

Question 28 (JAM 2021 • Q39 • ID: JAM_2021_Q39)

MSQ

+2 M

Neg: Nil (0)

Q. 39 Let V be a finite dimensional vector space and $T : V \rightarrow V$ be a linear transformation. Let $\mathcal{R}(T)$ denote the range of T and $\mathcal{N}(T)$ denote the null space $\{v \in V : Tv = 0\}$ of T . If $\text{rank}(T) = \text{rank}(T^2)$, then which of the following is/are necessarily true?

- (A) $\mathcal{N}(T) = \mathcal{N}(T^2)$. (B) $\mathcal{R}(T) = \mathcal{R}(T^2)$.
 (C) $\mathcal{N}(T) \cap \mathcal{R}(T) = \{0\}$. (D) $\mathcal{N}(T) = \{0\}$.

Question 29 (JAM 2021 • Q29 • ID: JAM_2021_Q29)

MCQ

+2 M

Neg: -0.67

JAM 2021

MATHEMATICS - MA

Q. 29 Let $n \geq 2$ be an integer. Let $A : \mathbb{C}^n \rightarrow \mathbb{C}^n$ be the linear transformation defined by

$$A(z_1, z_2, \dots, z_n) = (z_n, z_1, z_2, \dots, z_{n-1}).$$

Which one of the following statements is true for every $n \geq 2$?

- (A) A is nilpotent. (B) All eigenvalues of A are of modulus 1.
 (C) Every eigenvalue of A is either 0 or 1. (D) A is singular.

Question 30 (JAM 2021 • Q11 • ID: JAM_2021_Q11)

MCQ

+2 M

Neg: -0.67

Q. 11 – Q. 30 carry two marks each.

Q. 11 Let $M_n(\mathbb{R})$ be the real vector space of all $n \times n$ matrices with real entries, $n \geq 2$.

Let $A \in M_n(\mathbb{R})$. Consider the subspace W of $M_n(\mathbb{R})$ spanned by $\{I_n, A, A^2, \dots\}$. Then the dimension of W over $\mathbb{R}$ is necessarily

- (A) ∞ . (B) n^2 . (C) n . (D) at most n .

Question 31 (JAM 2021 • Q9 • ID: JAM_2021_Q9)

MCQ

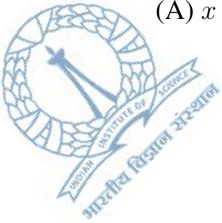
+1 M

Neg: -0.33

Q. 9 For an integer $k \geq 0$, let P_k denote the vector space of all real polynomials in one variable of degree less than or equal to k . Define a linear transformation $T : P_2 \rightarrow P_3$ by

$$Tf(x) = f''(x) + xf(x).$$

Which one of the following polynomials is not in the range of T ?

(A) $x + x^2$ (B) $x^2 + x^3 + 2$ (C) $x + x^3 + 2$ (D) $x + 1$ 

MA

3 / 17

Question 32 (JAM 2020 • Q59 • ID: JAM_2020_Q59)

NAT

+2 M

Neg: Nil (0)

Q. 59 Let $T : \mathbb{R}^7 \rightarrow \mathbb{R}^7$ be a linear transformation with $\text{Nullity}(T) = 2$. Then, the minimum possible value for $\text{Rank}(T^2)$ is _____.

Question 33 (JAM 2020 • Q49 • ID: JAM_2020_Q49)

NAT

+1 M

Neg: Nil (0)

Q. 49 Consider the real vector space $P_{2020} = \left\{ \sum_{i=0}^n a_i x^i : a_i \in \mathbb{R} \text{ and } 0 \leq n \leq 2020 \right\}$. Let W be the subspace given by

$$W = \left\{ \sum_{i=0}^n a_i x^i \in P_{2020} : a_i = 0 \text{ for all odd } i \right\}.$$

Then, the dimension of W is _____.

Question 34 (JAM 2020 • Q34 • ID: JAM_2020_Q34)

MSQ

+2 M

Neg: Nil (0)

JAM 2020

MATHEMATICS - MA

Q. 34 Let V be a non-zero vector space over a field F . Let $S \subset V$ be a non-empty set. Consider the following properties of S :

(I) For any vector space W over F , any map $f : S \rightarrow W$ extends to a linear map from V to W .

(II) For any vector space W over F and any two linear maps $f, g : V \rightarrow W$ satisfying $f(s) = g(s)$ for all $s \in S$, we have $f(v) = g(v)$ for all $v \in V$.

(III) S is linearly independent.

(IV) The span of S is V .

Which of the following statement(s) is /are true?

(A) (I) implies (IV)

(B) (I) implies (III)

(C) (II) implies (III)

(D) (II) implies (IV)

Question 35 (JAM 2020 • Q23 • ID: JAM_2020_Q23)

MCQ

+2 M

Neg: -0.67

Q. 23 Let M be a 4×3 real matrix and let $\{e_1, e_2, e_3\}$ be the standard basis of $\mathbb{R}^3$. Which of the following is true?

(A) If $\text{rank}(M) = 1$, then $\{Me_1, Me_2\}$ is a linearly independent set

(B) If $\text{rank}(M) = 2$, then $\{Me_1, Me_2\}$ is a linearly independent set

(C) If $\text{rank}(M) = 2$, then $\{Me_1, Me_3\}$ is a linearly independent set

(D) If $\text{rank}(M) = 3$, then $\{Me_1, Me_3\}$ is a linearly independent set

Question 36 (JAM 2020 • Q15 • ID: JAM_2020_Q15)

MCQ

+2 M

Neg: -0.67

Q. 15 Let $f(x, y) = \begin{cases} x^2 \sin \frac{1}{x}, & x \neq 0, y = 0 \\ y^2 \sin \frac{1}{y}, & y \neq 0, x = 0 \\ 0, & x = y = 0. \end{cases}$

Which of the following is true at $(0, 0)$?

- (A) f is not continuous
 (B) $\frac{\partial f}{\partial x}$ is continuous but $\frac{\partial f}{\partial y}$ is not continuous
 (C) f is not differentiable
 (D) f is differentiable but both $\frac{\partial f}{\partial x}$ and $\frac{\partial f}{\partial y}$ are not continuous

MA

6 / 17

Question 37 (JAM 2020 • Q4 • ID: JAM_2020_Q4)

MCQ

+1 M

Neg: -0.33

Q. 4 Let $g : \mathbb{R} \rightarrow \mathbb{R}$ be a twice differentiable function. If $f(x, y) = g(y) + xg'(y)$, then

- (A) $\frac{\partial f}{\partial x} + y \frac{\partial^2 f}{\partial x \partial y} = \frac{\partial f}{\partial y}$ (B) $\frac{\partial f}{\partial y} + y \frac{\partial^2 f}{\partial x \partial y} = \frac{\partial f}{\partial x}$
 (C) $\frac{\partial f}{\partial x} + x \frac{\partial^2 f}{\partial x \partial y} = \frac{\partial f}{\partial y}$ (D) $\frac{\partial f}{\partial y} + x \frac{\partial^2 f}{\partial x \partial y} = \frac{\partial f}{\partial x}$

MA

3 / 17

Question 38 (JAM 2019 • Q42 • ID: JAM_2019_Q42)

NAT

+1 M

Neg: Nil (0)

Q.42 Let W_1 and W_2 be subspaces of the real vector space $\mathbb{R}^{100}$ defined by

$$W_1 = \{ (x_1, x_2, \dots, x_{100}) : x_i = 0 \text{ if } i \text{ is divisible by } 4 \},$$

$$W_2 = \{ (x_1, x_2, \dots, x_{100}) : x_i = 0 \text{ if } i \text{ is divisible by } 5 \}.$$

Then the dimension of $W_1 \cap W_2$ is _____

Question 39 (JAM 2019 • Q33 • ID: JAM_2019_Q33)

MSQ

+2 M

Neg: Nil (0)

Q.33 Let S and T be linear transformations from a finite dimensional vector space V to itself such that $S(T(v)) = 0$ for all $v \in V$. Then

(A) $\text{rank}(T) \geq \text{nullity}(S)$

(B) $\text{rank}(S) \geq \text{nullity}(T)$

(C) $\text{rank}(T) \leq \text{nullity}(S)$

(D) $\text{rank}(S) \leq \text{nullity}(T)$

Question 40 (JAM 2019 • Q20 • ID: JAM_2019_Q20)

MCQ

+2 M

Neg: -0.67

Q.20 Let S be the family of orthogonal trajectories of the family of curves

$$2x^2 + y^2 = k, \text{ for } k \in \mathbb{R} \text{ and } k > 0.$$

If $\mathcal{C} \in S$ and $\mathcal{C}$ passes through the point $(1, 2)$, then $\mathcal{C}$ also passes through

(A) $(4, -\sqrt{2})$

(B) $(2, -4)$

(C) $(2, 2\sqrt{2})$

(D) $(4, 2\sqrt{2})$

Question 41 (JAM 2019 • Q4 • ID: JAM_2019_Q4)

MCQ

+1 M

Neg: -0.33

Q.4 Let $\{a_n\}_{n=0}^{\infty}$ and $\{b_n\}_{n=0}^{\infty}$ be sequences of positive real numbers such that $na_n < b_n < n^2a_n$ for all $n \geq 2$. If the radius of convergence of the power series $\sum_{n=0}^{\infty} a_n x^n$ is 4, then the power series $\sum_{n=0}^{\infty} b_n x^n$

(A) converges for all x with $|x| < 2$

(B) converges for all x with $|x| > 2$

(C) does not converge for any x with $|x| > 2$

(D) does not converge for any x with $|x| < 2$

Question 42 (JAM 2019 • Q1 • ID: JAM_2019_Q1)

MCQ

+1 M

Neg: -0.33

Q. 1 – Q.10 carry one mark each.Q.1 Let $a_1 = b_1 = 0$, and for each $n \geq 2$, let a_n and b_n be real numbers given by

$$a_n = \sum_{m=2}^n \frac{(-1)^m m}{(\log(m))^m} \text{ and } b_n = \sum_{m=2}^n \frac{1}{(\log(m))^m}.$$

Then which one of the following is TRUE about the sequences $\{a_n\}$ and $\{b_n\}$?

- (A) Both $\{a_n\}$ and $\{b_n\}$ are divergent
- (B) $\{a_n\}$ is convergent and $\{b_n\}$ is divergent
- (C) $\{a_n\}$ is divergent and $\{b_n\}$ is convergent
- (D) Both $\{a_n\}$ and $\{b_n\}$ are convergent

Question 43 (JAM 2018 • Q57 • ID: JAM_2018_Q57)

NAT

+2 M

Neg: Nil (0)

Q.57 Suppose $Q \in M_{3 \times 3}(\mathbb{R})$ is a matrix of rank 2. Let $T: M_{3 \times 3}(\mathbb{R}) \rightarrow M_{3 \times 3}(\mathbb{R})$ be the linear transformation defined by $T(P) = QP$. Then the rank of T is _____

Question 44 (JAM 2018 • Q49 • ID: JAM_2018_Q49)

NAT

+1 M

Neg: Nil (0)

Q.49 Let W_1 be the real vector space of all 5×2 matrices such that the sum of the entries in each row is zero. Let W_2 be the real vector space of all 5×2 matrices such that the sum of the entries in each column is zero. Then the dimension of the space $W_1 \cap W_2$ is _____

Question 45 (JAM 2018 • Q26 • ID: JAM_2018_Q26)

MCQ

+2 M

Neg: -0.67

Q.26 Let H be the quotient group $\mathbb{Q}/\mathbb{Z}$. Consider the following statements.

- I. Every cyclic subgroup of H is finite.
- II. Every finite cyclic group is isomorphic to a subgroup of H .

Which one of the following holds?

- (A) I is TRUE but II is FALSE
- (B) II is TRUE but I is FALSE
- (C) both I and II are TRUE
- (D) neither I nor II is TRUE

Question 46 (JAM 2018 • Q16 • ID: JAM_2018_Q16)

MCQ

+2 M

Neg: -0.67

Q.16 For $x \in \mathbb{R}$, let $f(x) = \begin{cases} x^3 \sin\left(\frac{1}{x}\right), & x \neq 0 \\ 0, & x = 0 \end{cases}$. Then which one of the following is FALSE?

- (A) $\lim_{x \rightarrow 0} \frac{f(x)}{x} = 0$
- (B) $\lim_{x \rightarrow 0} \frac{f(x)}{x^2} = 0$
- (C) $\frac{f(x)}{x^2}$ has infinitely many maxima and minima on the interval $(0,1)$
- (D) $\frac{f(x)}{x^4}$ is continuous at $x = 0$ but not differentiable at $x = 0$

Question 47 (JAM 2018 • Q6 • ID: JAM_2018_Q6)

MCQ

+1 M

Neg: -0.33

Q.6 In $\mathbb{R}^3$, the cosine of the acute angle between the surfaces $x^2 + y^2 + z^2 - 9 = 0$ and $z - x^2 - y^2 + 3 = 0$ at the point $(2, 1, 2)$ is

- (A) $\frac{8}{5\sqrt{21}}$
- (B) $\frac{10}{5\sqrt{21}}$
- (C) $\frac{8}{3\sqrt{21}}$
- (D) $\frac{10}{3\sqrt{21}}$

Question 48 (JAM 2017 • Q46 • ID: JAM_2017_Q46)

NAT

+1 M

Neg: Nil (0)

Q.46 Let $v_1 = \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}$ and $v_2 = \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix}$. Let M be the matrix whose columns are $v_1, v_2, 2v_1 - v_2, v_1 + 2v_2$ in that order. Then the number of linearly independent solutions of the homogeneous system of linear equations $Mx = 0$ is _____

Question 49 (JAM 2017 • Q22 • ID: JAM_2017_Q22)

MCQ

+2 M

Neg: -0.67

Q.22 Let $\mathcal{P}_3$ denote the real vector space of all polynomials with real coefficients of degree at most 3. Consider the map $T: \mathcal{P}_3 \rightarrow \mathcal{P}_3$ given by $T(p(x)) = p''(x) + p(x)$. Then

(A) T is neither one-one nor onto

(B) T is both one-one and onto

(C) T is one-one but not onto

(D) T is onto but not one-one

Question 50 (JAM 2016 • Q59 • ID: JAM_2016_Q59)

NAT

+2 M

Neg: Nil (0)

Q.59 Let $A = \begin{bmatrix} 1 & 2 & 0 \\ -1 & 5 & 2 \end{bmatrix}$, $B = \begin{bmatrix} 1 & 2 \\ -1 & 0 \\ 3 & 1 \end{bmatrix}$, $N(A)$ the null space of A and $R(B)$ the range space of B . Then the dimension of $N(A) \cap R(B)$ over $\mathbb{R}$ is _____

Question 51 (JAM 2016 • Q39 • ID: JAM_2016_Q39)

MSQ

+2 M

Neg: Nil (0)

Q.39 Consider the set $V = \left\{ \begin{bmatrix} x \\ y \\ z \end{bmatrix} \in \mathbb{R}^3 \mid \alpha x + \beta y + z = \gamma, \alpha, \beta, \gamma \in \mathbb{R} \right\}$. For which of the following choice(s) the set V becomes a two dimensional subspace of $\mathbb{R}^3$ over $\mathbb{R}$?

(A) $\alpha = 0, \beta = 1, \gamma = 0$

(B) $\alpha = 0, \beta = 1, \gamma = 1$

(C) $\alpha = 1, \beta = 0, \gamma = 0$

(D) $\alpha = 1, \beta = 1, \gamma = 0$

Question 52 (JAM 2016 • Q27 • ID: JAM_2016_Q27)

MCQ

+2 M

Neg: -0.67

Q.27 Let $M = \begin{bmatrix} 0 & 1 & -2 \\ -1 & 0 & \alpha \\ 2 & -\alpha & 0 \end{bmatrix}$, $\alpha \in \mathbb{R} \setminus \{0\}$ and $\mathbf{b}$ a non-zero vector such that $M\mathbf{x} = \mathbf{b}$ for some $\mathbf{x} \in \mathbb{R}^3$. Then the value of $\mathbf{x}^T \mathbf{b}$ is

- (A) $-\alpha$ (B) α (C) 0 (D) 1

Question 53 (JAM 2016 • Q17 • ID: JAM_2016_Q17)

MCQ

+2 M

Neg: -0.67

Q.17 Let $f: \mathbb{R}^2 \rightarrow \mathbb{R}$ be defined by

$$f(x, y) = \begin{cases} \frac{xy^2}{x+y} & \text{if } x+y \neq 0 \\ 0 & \text{if } x+y = 0 \end{cases}.$$

Then the value of $\left(\frac{\partial^2 f}{\partial x \partial y} + \frac{\partial^2 f}{\partial y \partial x} \right)$ at the point $(0, 0)$ is

- (A) 0 (B) 1 (C) 2 (D) 4

Question 54 (JAM 2016 • Q2 • ID: JAM_2016_Q2)

MCQ

+1 M

Neg: -0.33

Q.2 Let P be the vector space (over $\mathbb{R}$) of all polynomials of degree ≤ 3 with real coefficients. Consider the linear transformation $T: P \rightarrow P$ defined by

$$T(a_0 + a_1x + a_2x^2 + a_3x^3) = a_3 + a_2x + a_1x^2 + a_0x^3.$$

Then the matrix representation M of T with respect to the ordered basis $\{1, x, x^2, x^3\}$ satisfies

- (A) $M^2 + I_4 = 0$ (B) $M^2 - I_4 = 0$
(C) $M - I_4 = 0$ (D) $M + I_4 = 0$

Question 55 (JAM 2015 • Q54 • ID: JAM_2015_Q54)

NAT

+2 M

Neg: Nil (0)

Q.14 Let $M_2(\mathbb{R})$ be the vector space of 2×2 real matrices. Let V be a subspace of $M_2(\mathbb{R})$ defined by

$$V = \left\{ A \in M_2(\mathbb{R}) : A \begin{bmatrix} 0 & 2 \\ 3 & 1 \end{bmatrix} = \begin{bmatrix} 0 & 2 \\ 3 & 1 \end{bmatrix} A \right\}$$

Then the dimension of V is _____

MA

15/16

Question 56 (JAM 2015 • Q48 • ID: JAM_2015_Q48)

NAT

+1 M

Neg: Nil (0)

Q.8

If the set $\left\{ \begin{bmatrix} x & -x \\ -1 & 0 \end{bmatrix}, \begin{bmatrix} 0 & -1 \\ x & x \end{bmatrix}, \begin{bmatrix} 1 & -1 \\ 1 & 0 \end{bmatrix} \right\}$ is linearly dependent in the vector space of all 2×2 matrices with real entries, then x is equal to _____

Question 57 (JAM 2015 • Q35 • ID: JAM_2015_Q35)

MSQ

+2 M

Neg: Nil (0)

Q.5

Let V be the set of 2×2 matrices $\begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix}$ with complex entries such that $a_{11} + a_{22} = 0$. Let W be the set of matrices in V with $a_{12} + \overline{a_{21}} = 0$. Then, under usual matrix addition and scalar multiplication, which of the following is (are) true?

- (A) V is a vector space over $\mathbb{C}$
- (B) W is a vector space over $\mathbb{C}$
- (C) V is a vector space over $\mathbb{R}$
- (D) W is a vector space over $\mathbb{R}$

Question 58 (JAM 2015 • Q17 • ID: JAM_2015_Q17)

MCQ

+2 M

Neg: -0.67

Q.17 The limit

$$\lim_{x \rightarrow 0^+} \frac{1}{\sin^2 x} \int_{\frac{x}{2}}^x \sin^{-1} t \, dt$$

is equal to

- (A) 0 (B) $\frac{1}{8}$ (C) $\frac{1}{4}$ (D) $\frac{3}{8}$

Question 59 (JAM 2015 • Q4 • ID: JAM_2015_Q4)

MCQ

+1 M

Neg: -0.33

Q.4 Let S be a nonempty subset of $\mathbb{R}$. If S is a finite union of disjoint bounded intervals, then which one of the following is true?

- (A) If S is not compact, then $\sup S \notin S$ and $\inf S \notin S$
 (B) Even if $\sup S \in S$ and $\inf S \in S$, S need not be compact
 (C) If $\sup S \in S$ and $\inf S \in S$, then S is compact
 (D) Even if S is compact, it is not necessary that $\sup S \in S$ and $\inf S \in S$

Question 60 (JAM 2014 • Q13 • ID: JAM_2014_Q13)

MCQ

+2 M

Neg: -0.67

Q.13 Consider the subspace $W = \{(x_1, x_2, \dots, x_{10}) \in \mathbb{R}^{10} : x_n = x_{n-1} + x_{n-2} \text{ for } 3 \leq n \leq 10\}$ of the vector space $\mathbb{R}^{10}$. The dimension of W is

- (A) 2 (B) 3 (C) 9 (D) 10

Question 61 (JAM 2014 • Q9 • ID: JAM_2014_Q9)

MCQ

+1 M

Neg: -0.33

Q.9 Let $T: \mathbb{R}^3 \rightarrow \mathbb{R}^3$ be the linear transformation defined by $T(x, y, z) = (x + y, y + z, z + x)$ for all $(x, y, z) \in \mathbb{R}^3$. Then

- (A) $\text{rank}(T) = 0$, $\text{nullity}(T) = 3$ (B) $\text{rank}(T) = 2$, $\text{nullity}(T) = 1$
 (C) $\text{rank}(T) = 1$, $\text{nullity}(T) = 2$ (D) $\text{rank}(T) = 3$, $\text{nullity}(T) = 0$

Question 62 (JAM 2014 • Q8 • ID: JAM_2014_Q8)

MCQ

+1 M

Neg: -0.33

Q.8 Which one of the following is a subspace of the vector space $\mathbb{R}^3$?

- (A) $\{(x, y, z) \in \mathbb{R}^3 : x + 2y = 0, 2x + 3z = 0\}$
- (B) $\{(x, y, z) \in \mathbb{R}^3 : 2x + 3y + 4z - 3 = 0, z = 0\}$
- (C) $\{(x, y, z) \in \mathbb{R}^3 : x \geq 0, y \geq 0\}$
- (D) $\{(x, y, z) \in \mathbb{R}^3 : x - 1 = 0, y = 0\}$

Question 63 (JAM 2013 • Q10 • ID: JAM_2013_Q10)

MCQ

+2 M

Neg: -0.67

Q.10 Let $\mathcal{P}_n$ be the real vector space of all polynomials of degree at most n . Let $D : \mathcal{P}_n \rightarrow \mathcal{P}_{n-1}$ and $T : \mathcal{P}_n \rightarrow \mathcal{P}_{n+1}$ be the linear transformations defined by

$$D(a_0 + a_1x + a_2x^2 + \dots + a_nx^n) = a_1 + 2a_2x + \dots + na_nx^{n-1},$$

$$T(a_0 + a_1x + a_2x^2 + \dots + a_nx^n) = a_0x + a_1x^2 + a_2x^3 + \dots + a_nx^{n+1},$$

respectively. If A is the matrix representation of the transformation $DT - TD : \mathcal{P}_n \rightarrow \mathcal{P}_n$ with respect to the standard basis of $\mathcal{P}_n$, then the trace of A is

- (A) $-n$ (B) n (C) $n+1$ (D) $-(n+1)$

Question 64 (JAM 2013 • Q9 • ID: JAM_2013_Q9)

MCQ

+2 M

Neg: -0.67

Q.9 Let V be the vector space of all 2×2 matrices over $\mathbb{R}$. Consider the subspaces

$$W_1 = \left\{ \begin{pmatrix} a & -a \\ c & d \end{pmatrix} : a, c, d \in \mathbb{R} \right\} \text{ and } W_2 = \left\{ \begin{pmatrix} a & b \\ -a & d \end{pmatrix} : a, b, d \in \mathbb{R} \right\}.$$

If $m = \dim(W_1 \cap W_2)$ and $n = \dim(W_1 + W_2)$, then the pair (m, n) is

- (A) $(2, 3)$ (B) $(2, 4)$ (C) $(3, 4)$ (D) $(1, 3)$

Question 65 (JAM 2013 • Q1 • ID: JAM_2013_Q1)

MCQ

+2 M

Neg: -0.67

Q.1 Let $A = \begin{pmatrix} 1 & 1 & 1 \\ 3 & -1 & 1 \\ 1 & 5 & 3 \end{pmatrix}$ and V be the vector space of all $X \in \mathbb{R}^3$ such that $AX = 0$. Then $\dim(V)$ is

(A) 0 (B) 1 (C) 2 (D) 3

Question 66 (JAM 2012 • Q11 • ID: JAM_2012_Q11)

MCQ

+6 M

Neg: -2.0

Q.11 Consider the following subspace of $\mathbb{R}^3$:

$$W = \{(x, y, z) \in \mathbb{R}^3 \mid 2x + 2y + z = 0, 3x + 3y - 2z = 0, x + y - 3z = 0\}.$$

The dimension of W is

(A) 0 (B) 1 (C) 2 (D) 3

Question 67 (JAM 2012 • Q10 • ID: JAM_2012_Q10)

MCQ

+6 M

Neg: -2.0

Q.10 Let W be a vector space over $\mathbb{R}$ and let $T: \mathbb{R}^6 \rightarrow W$ be a linear transformation such that $S = \{Te_2, Te_4, Te_6\}$ spans W . Which one of the following must be TRUE?

- (A) S is a basis of W
- (B) $T(\mathbb{R}^6) \neq W$
- (C) $\{Te_1, Te_3, Te_5\}$ spans W
- (D) $\ker(T)$ contains more than one element

Question 68 (JAM 2011 • Q13 • ID: JAM_2011_Q13)

MCQ

+6 M

Neg: -2.0

Q.13 For $n \neq m$, let $T_1: \mathbb{R}^n \rightarrow \mathbb{R}^m$ and $T_2: \mathbb{R}^m \rightarrow \mathbb{R}^n$ be linear transformations such that $T_1 T_2$ is bijective. Then

- (A) $\text{rank}(T_1) = n$ and $\text{rank}(T_2) = m$ (B) $\text{rank}(T_1) = m$ and $\text{rank}(T_2) = n$
 (C) $\text{rank}(T_1) = n$ and $\text{rank}(T_2) = n$ (D) $\text{rank}(T_1) = m$ and $\text{rank}(T_2) = m$

Question 69 (JAM 2011 • Q12 • ID: JAM_2011_Q12)

MCQ

+6 M

Neg: -2.0

Q.12 Let $T: \mathbb{R}^n \rightarrow \mathbb{R}^n$ be a linear transformation, where $n \geq 2$. For $k \leq n$, let

$$E = \{v_1, v_2, \dots, v_k\} \subseteq \mathbb{R}^n \text{ and } F = \{Tv_1, Tv_2, \dots, Tv_k\}.$$

Then

- (A) If E is linearly independent, then F is linearly independent
 (B) If F is linearly independent, then E is linearly independent
 (C) If E is linearly independent, then F is linearly dependent
 (D) If F is linearly independent, then E is linearly dependent

Question 70 (JAM 2010 • Q15 • ID: JAM_2010_Q15)

MCQ

+6 M

Neg: -2.0

Q.15 Let $T: \mathbb{R}^3 \rightarrow \mathbb{R}^3$ be the linear transformation whose matrix with respect to the standard basis of $\mathbb{R}^3$ is $\begin{pmatrix} 0 & a & b \\ -a & 0 & c \\ -b & -c & 0 \end{pmatrix}$, where a, b, c are real numbers not all zero. Then T

- (A) is one-to-one
 (B) is onto
 (C) does not map any line through the origin onto itself
 (D) has rank 1

Question 71 (JAM 2010 • Q14 • ID: JAM_2010_Q14)

MCQ

+6 M

Neg: -2.0

Q.14 Let $T: \mathbf{R}^3 \rightarrow \mathbf{R}^3$ be the linear transformation whose matrix with respect to the standard basis $\{e_1, e_2, e_3\}$ of $\mathbf{R}^3$ is $\begin{pmatrix} 0 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 0 \end{pmatrix}$. Then T

- (A) maps the subspace spanned by e_1 and e_2 into itself
- (B) has distinct eigenvalues
- (C) has eigenvectors that span $\mathbf{R}^3$
- (D) has a non-zero null space

Question 72 (JAM 2010 • Q11 • ID: JAM_2010_Q11)

MCQ

+6 M

Neg: -2.0

Q.11 Let $T: \mathbf{R}^2 \rightarrow \mathbf{R}^2$ be a linear transformation such that $T((1, 2)) = (2, 3)$ and $T((0, 1)) = (1, 4)$. Then $T((5, 6))$ is

- (A) $(6, -1)$
- (B) $(-6, 1)$
- (C) $(-1, 6)$
- (D) $(1, -6)$

Question 73 (JAM 2009 • Q4 • ID: JAM_2009_Q4)

MCQ

+6 M

Neg: -2.0

Q.4 Let $T: \mathbf{R}^4 \rightarrow \mathbf{R}^4$ be a linear transformation satisfying $T^3 + 3T^2 = 4I$, where I is the identity transformation. Then the linear transformation $S = T^4 + 3T^3 - 4I$ is

- (A) one-one but not onto
- (B) onto but not one-one
- (C) invertible
- (D) non-invertible

Question 74 (JAM 2009 • Q1 • ID: JAM_2009_Q1)

MCQ

+6 M

Neg: -2.0

Q.1 Let V be the vector space of all 6×6 real matrices over the field $\mathbf{R}$. Then the dimension of the subspace of V consisting of all symmetric matrices is

- (A) 15
- (B) 18
- (C) 21
- (D) 35

Question 75 (JAM 2008 • Q11 • ID: JAM_2008_Q11)

MCQ

+6 M

Neg: -2.0

Q.11 Let $T : \mathbb{R}^3 \rightarrow \mathbb{R}^3$ be a linear transformation such that $T(1, 2, 3) = (1, 2, 3)$, $T(1, 5, 0) = (2, 10, 0)$ and $T(-1, 2, -1) = (-3, 6, -3)$. The dimension of the vector space spanned by all the eigenvectors of T is

(A) 0

(B) 1

(C) 2

(D) 3

Question 76 (JAM 2007 • Q5 • ID: JAM_2007_Q5)

MCQ

+6 M

Neg: -2.0

5. Which of the following sets is a basis for the subspace

$$W = \left\{ \begin{bmatrix} x & y \\ 0 & t \end{bmatrix} : x + 2y + t = 0, y + t = 0 \right\}$$

of the vector space of all real 2×2 matrices?

(A) $\left\{ \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix} \right\}$

(B) $\left\{ \begin{bmatrix} 2 & 1 \\ 0 & -1 \end{bmatrix}, \begin{bmatrix} 1 & -1 \\ 0 & 1 \end{bmatrix} \right\}$

(C) $\left\{ \begin{bmatrix} -1 & 1 \\ 2 & -1 \end{bmatrix} \right\}$

(D) $\left\{ \begin{bmatrix} 1 & -1 \\ 0 & 1 \end{bmatrix} \right\}$

Question 77 (JAM 2006 • Q8 • ID: JAM_2006_Q8)

MCQ

+6 M

Neg: -2.0

8. Let $T : \mathbb{R}^3 \rightarrow \mathbb{R}^3$ be defined by

$$T(x_1, x_2, x_3) = (x_1 - x_2, x_1 - x_2, 0).$$

If $N(T)$ and $R(T)$ denote the null space and the range space of T respectively, then

(A) $\dim N(T) = 2$

(B) $\dim R(T) = 2$

(C) $R(T) = N(T)$

(D) $N(T) \subset R(T)$

Question 78 (JAM 2005 • Q7 • ID: JAM_2005_Q7)

MCQ

+6 M

Neg: -2.0

7. Let $T : \mathbf{R}^3 \longrightarrow \mathbf{R}^3$ be a linear transformation and I be the identity transformation of $\mathbf{R}^3$. If there is a scalar c and a non-zero vector $x \in \mathbf{R}^3$ such that $T(x) = cx$, then $\text{rank}(T - cI)$
- (A) cannot be 0
 - (B) cannot be 1
 - (C) cannot be 2
 - (D) cannot be 3

OFFICIAL MASTER ANSWER KEY & EVALUATION RUBRIC

Finite Dimensional Vector Spaces (2005–2026 Archive) • All Official Answers & Acceptance Ranges

Q. No.	Type	Marks	Official Key	Q. No.	Type	Marks	Official Key	Q. No.	Type	Marks	Official Key
1	MSQ	+2	B;D	27	NAT	+1	5	53	MCQ	+2	B
2	MCQ	+2	D	28	MSQ	+2	A, B, C	54	MCQ	+1	B
3	MCQ	+2	C	29	MCQ	+2	B	55	NAT	+2	2
4	NAT	+2	1 to 1	30	MCQ	+2	D	56	NAT	+1	−1
5	NAT	+1	1 to 1	31	MCQ	+1	D	57	MSQ	+2	A;C;D
6	NAT	+1	3.98 to 4.02	32	NAT	+2	3 to 3	58	MCQ	+2	D
7	NAT	+1	2 to 2	33	NAT	+1	1011 to 1011	59	MCQ	+1	B
8	MSQ	+2	C;D	34	MSQ	+2	B;D	60	MCQ	+2	A
9	MSQ	+2	A;B	35	MCQ	+2	D	61	MCQ	+1	D
10	MCQ	+2	C	36	MCQ	+2	D	62	MCQ	+1	A
11	MCQ	+1	C	37	MCQ	+1	C	63	MCQ	+2	C
12	MCQ	+1	D	38	NAT	+1	60 to 60	64	MCQ	+2	B
13	NAT	+2	5 to 5	39	MSQ	+2	C, D	65	MCQ	+2	B
14	NAT	+1	14.95 to 15.05	40	MCQ	+2	C	66	MCQ	+6	B
15	NAT	+1	6 to 6	41	MCQ	+1	A	67	MCQ	+6	D
16	MCQ	+2	D	42	MCQ	+1	D	68	MCQ	+6	D
17	MCQ	+2	B	43	NAT	+2	6 to 6	69	MCQ	+6	B
18	MCQ	+2	A	44	NAT	+1	4 to 4	70	MCQ	+6	C
19	NAT	+2	4 to 4	45	MCQ	+2	C	71	MCQ	+6	C
20	NAT	+1	3 to 3	46	MCQ	+2	D	72	MCQ	+6	A
21	MSQ	+2	B, D	47	MCQ	+1	C	73	MCQ	+6	D
22	MSQ	+2	A, B, C	48	NAT	+1	1.9 to 2.1	74	MCQ	+6	C
23	MCQ	+2	D	49	MCQ	+2	B	75	MCQ	+6	D
24	MCQ	+2	C	50	NAT	+2	1.0 to 1.0	76	MCQ	+6	D
25	MCQ	+1	A	51	MSQ	+2	A;C;D	77	MCQ	+6	A
26	NAT	+2	8	52	MCQ	+2	C	78	MCQ	+6	D

SCORING METHODOLOGY & EVALUATION GUIDELINES

- **Multiple Choice Questions (MCQ):** Full marks for correct single choice. Negative marks deducted for each incorrect attempt (−1/3 for 1-mark questions, −2/3 for 2-mark questions). Unattempted questions award zero.
- **Multiple Select Questions (MSQ):** Full marks if and only if all correct choices are marked and no incorrect choice is selected. Zero marks for partial selection or any incorrect choice. No negative marking.
- **Numerical Answer Type (NAT):** Any numerical value falling strictly within the specified official range (e.g., [a, b]) receives full credit. No negative marking.
- **Marks to All (MTA):** Questions officially designated as MTA award full marks to all candidates regardless of attempt.